

Experiment 3: Spectrum Analysis of Sub-1 GHz, Mid-band & mmWave by Plotting Propagation Loss

Objectives, MATLAB Programs and Output Images

Objective 1

To compute the Propagation Loss (dB) for Sub-1 GHz, Mid-band, and mmWave frequency ranges using standard path loss models.

MATLAB Program

```
clc; clear; close all;

d = 0.1:0.1:5;          % Distance (km)
f = [900 3500 28000];    % Frequency (MHz)

PL1 = 32.44 + 20*log10(f(1)) + 20*log10(d);
PL2 = 32.44 + 20*log10(f(2)) + 20*log10(d);
PL3 = 32.44 + 20*log10(f(3)) + 20*log10(d);

figure

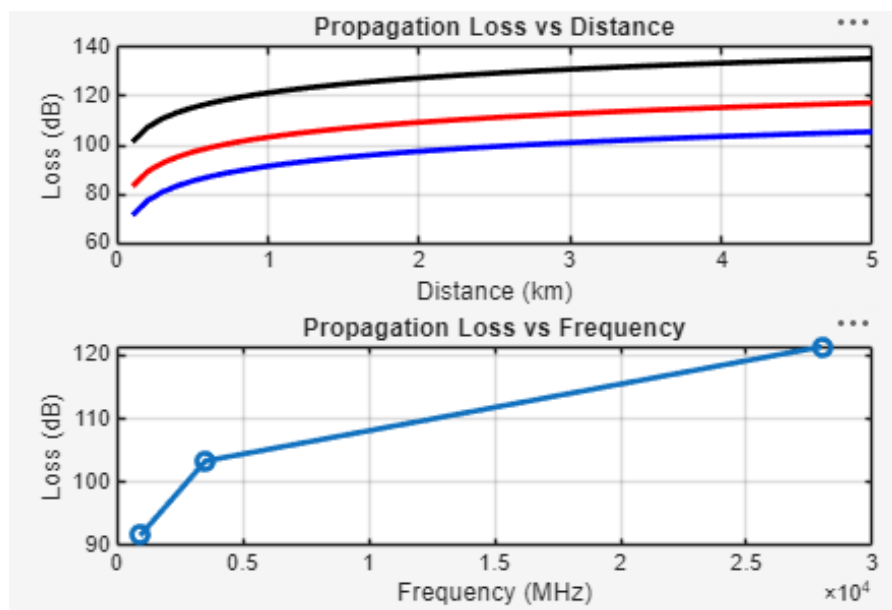
% ----- Graph 1 -----
subplot(2,1,1)

plot(d,PL1,'b','LineWidth',2); hold on
plot(d,PL2,'r','LineWidth',2)
plot(d,PL3,'k','LineWidth',2)
title('Propagation Loss vs Distance')
xlabel('Distance (km)')
ylabel('Loss (dB)')
grid on

% ----- Graph 2 (NO legend used to avoid error) -----
subplot(2,1,2)
Freq = [900 3500 28000];
Loss = [PL1(10) PL2(10) PL3(10)];

plot(Freq,Loss,'-o','LineWidth',2)
title('Propagation Loss vs Frequency')
xlabel('Frequency (MHz)')
ylabel('Loss (dB)')
grid on
```

Output Image – Objective 1



Objective 2

To evaluate the impact of different Operating Frequencies (GHz) on signal attenuation and coverage performance.

MATLAB Program

```
clc; clear; close all;

d = 2; % fixed distance (km)

f = [900 3500 28000]; % MHz

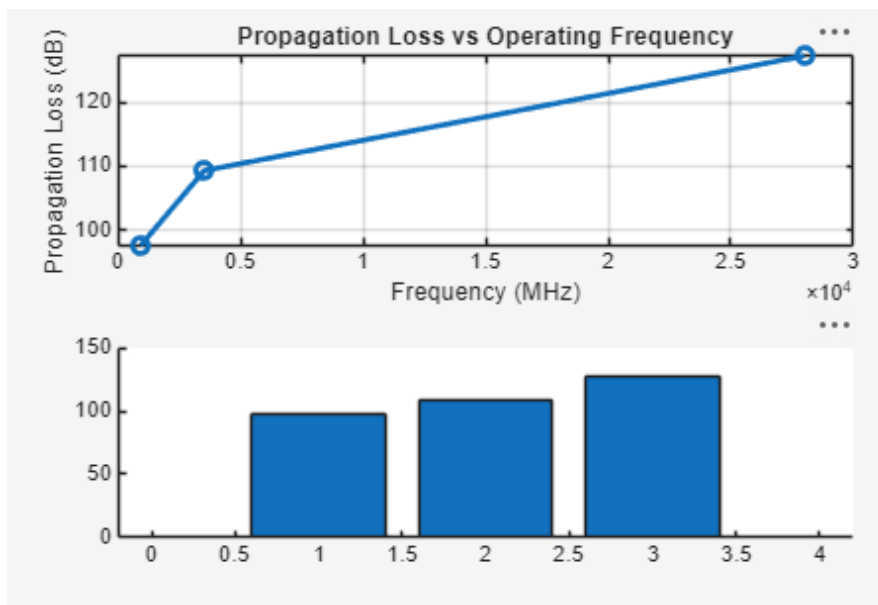
PL = 32.44 + 20*log10(f) + 20*log10(d);

figure

% ----- Graph 1: Frequency vs Loss -----
subplot(2,1,1)
plot(f,PL,'-o','LineWidth',2)
title('Propagation Loss vs Operating Frequency')
xlabel('Frequency (MHz)')
ylabel('Propagation Loss (dB)')
grid on

% ----- Graph 2: Band Comparison -----
subplot(2,1,2)
bar(PL)
title('Comparison of Propagation Loss Across Bands')
xlabel('Frequency Bands (Index: 1=Sub-1GHz, 2=Mid-band, 3=mmWave)')
ylabel('Propagation Loss (dB)')
xticklabels({'Sub-1GHz','Mid-band','mmWave'})
grid on
```

Output Image – Objective 2



Objective 3

To analyze how Communication Distance (km) affects propagation loss across different spectrum bands.

MATLAB Program

```
clc; clear; close all;

% Frequency bands (MHz)
f = [900 3500 28000];

% Distance (km)
d = 2;

% Propagation Loss (FSPL)
PL = 32.44 + 20*log10(f) + 20*log10(d);

figure

% ----- Graph 1: Simple Line Plot (SAFE) -----
subplot(2,1,1)
plot(f, PL, '-o', 'LineWidth', 2)
title('Propagation Loss vs Operating Frequency')
xlabel('Frequency (MHz)')
ylabel('Propagation Loss (dB)')
grid on

% ----- Graph 2: Safe Comparison Plot -----
subplot(2,1,2)
plot(1:3, PL, '-s', 'LineWidth', 2)
title('Band-wise Propagation Loss Comparison')
xlabel('Band Index (1=Sub1GHz, 2=Mid-band, 3=mmWave)')
ylabel('Propagation Loss (dB)')
grid on
```

Output Image – Objective 3

